

CE 310

Week 8 — Midterm

University of Arizona · Dr. Wooyoung Jung · Fall 2026

This Week — Three Sessions

Mon Oct 12 (Session 1): Review — connections across Weeks 1–7; common exam topics; Q&A. Nothing to turn in.

Wed Oct 14 (Session 2) — Concept Exam

- **Duration:** 50 minutes
- **Points:** 50 pts
- **Allowed:** Pencil + scientific calculator
- **Not allowed:** Computer, phone, internet
- Submit your exam paper before leaving

Fri Oct 16 (Session 3) — Lab Practicum

- **Duration:** 50 minutes
- **Points:** 100 pts
- **Allowed:** Excel, personal notes, calculator
- **Not allowed:** Internet access
- **Bring:** Laptop + dataset file (download now)

Wed Oct 14 — Concept Exam Format

Part	Type	Count	Points
1	Multiple Choice	10 questions × 2 pts	20 pts
2	Short Answer	5 questions × 4 pts	20 pts
3	By-Hand Calculation	2 questions × 5 pts	10 pts
		Total	50 pts

- One MC question per week (W1–W7) — the remaining three draw from W5 and W7
- Part 3 requires **visible work** — partial credit requires shown arithmetic
- Calculator permitted for z-score and covariance calculations in Part 3

Topics Covered — Weeks 1–7

Week	Topic	Key Skill
1	Flaw of Averages · Design Percentiles	LARGE, PERCENTILE
2	Descriptive Stats · Spread & Shape	STDEV, SKEW, IQR
3	Conditional Probability · Pivot Tables	COUNTIFS, $P(A B)$
4	Return Periods · Poisson Distribution	$P(X=k)$ by hand, exceedance
5	Normal Distribution	NORM.DIST, NORM.INV, z-score
6	Empirical Percentiles · Duration Curves	LARGE(k), COUNTIF exceedance
7	Correlation · r^2	CORREL, r^2 , Simpson's Paradox

Fri Oct 16 — Lab Practicum Format

Dataset: Denver boutique hotel · 365 daily rows · `Week8_Practicum_Dataset.csv`

Section	Content	Points
A	Descriptive Stats & Normal Model (8 questions)	40 pts
B	Conditional Analysis (6 questions)	30 pts
Chart	Scatter plot — Outdoor Temp vs. HVAC Energy	15 pts
Written	Three interpretation sentences	15 pts
	Total	100 pts

⚠ Download `Week8_Practicum_Dataset.csv` **before class** — no internet access during the practicum.

You're Ready

Weeks 1–7 built every skill on this exam.

Trust your work. Show your steps. Check your formulas.

Good luck.